

**Assigned group: C**

**Subject: 1**

## Instructions

Your task is to predict the reference classifier's class assignment for each time-series plot. There is no feedback during the survey.

The classifier outputs are shown as Red and Blue. These colour names are only response labels for this study.

For every test plot, choose Red or Blue and then rate how confident you are from 1 (very low) to 5 (very high).

### Short Background Form

Please answer on a 1-5 scale.

How familiar are you with machine learning classification?

☐ 1 Not familiar   ☐ 2 Slightly familiar   ☐ 3 Moderately familiar   ☐ 4 Familiar   ☐ 5 Very familiar

How familiar are you with reading time-series plots?

☐ 1 Not familiar   ☐ 2 Slightly familiar   ☐ 3 Moderately familiar   ☐ 4 Familiar   ☐ 5 Very familiar

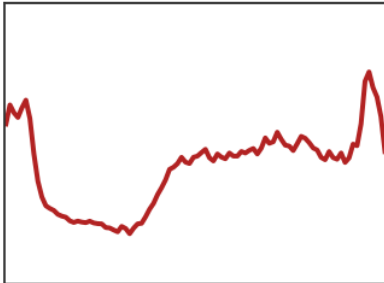
How familiar are you with explainable AI or rule-based explanations?

☐ 1 Not familiar   ☐ 2 Slightly familiar   ☐ 3 Moderately familiar   ☐ 4 Familiar   ☐ 5 Very familiar

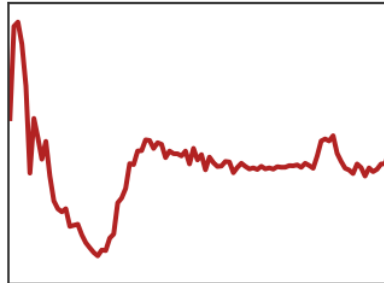


Study the example plots below. Then classify the 10 test instances on the next pages.

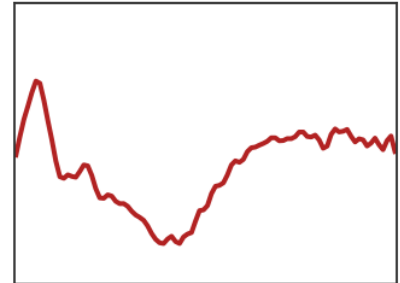
## Red reference examples



Example 1



Example 2

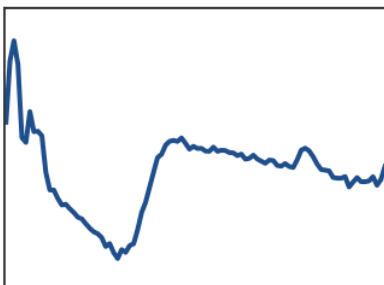


Example 3

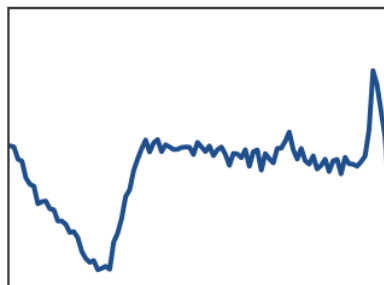
### Reference notes

Use the three example plots above as your reference for the Red class.

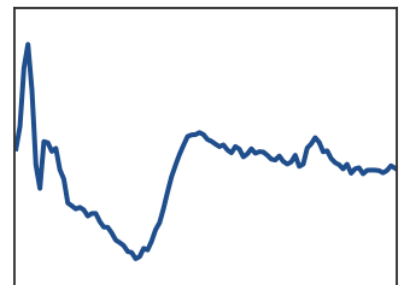
## Blue reference examples



Example 1



Example 2

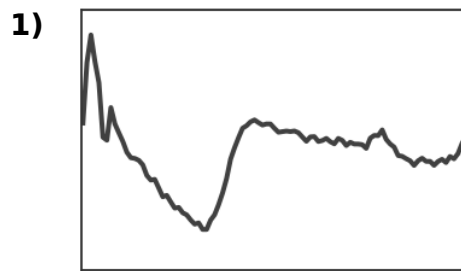


Example 3

### Reference notes

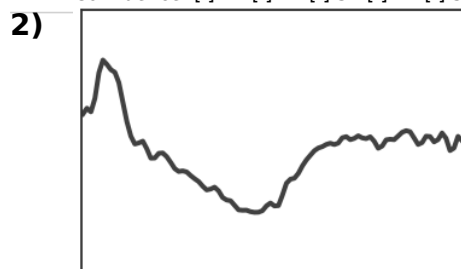
Use the three example plots above as your reference for the Blue class.

For each plot, choose Red or Blue and then rate your confidence from 1 (very low) to 5 (very high).



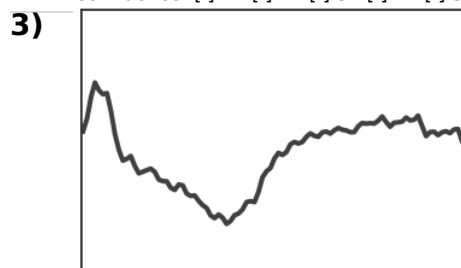
Label: ☐ Red ☐ Blue

Confidence: ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5



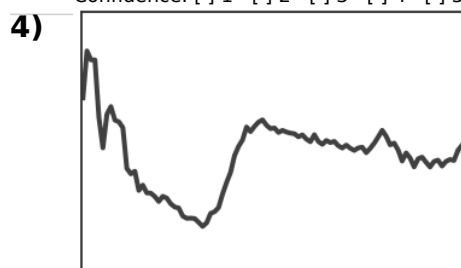
Label: ☐ Red ☐ Blue

Confidence: ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5



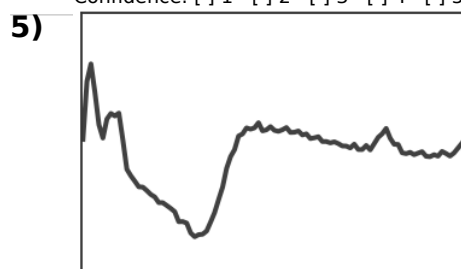
Label: ☐ Red ☐ Blue

Confidence: ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5



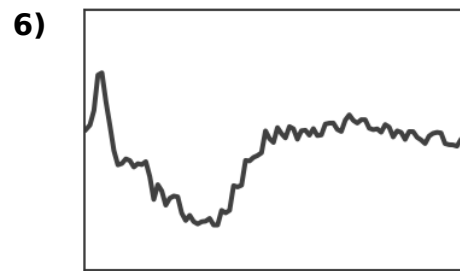
Label: ☐ Red ☐ Blue

Confidence: ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5



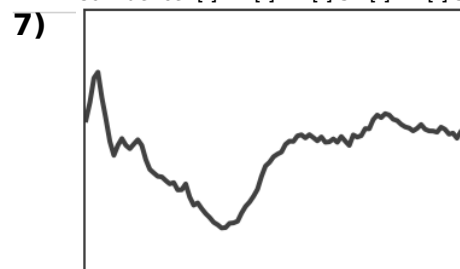
Label: ☐ Red ☐ Blue

Confidence: ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5



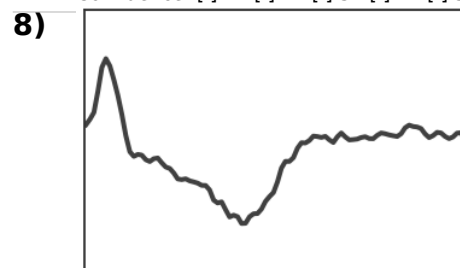
Label: ☐ Red ☐ Blue

Confidence: ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5



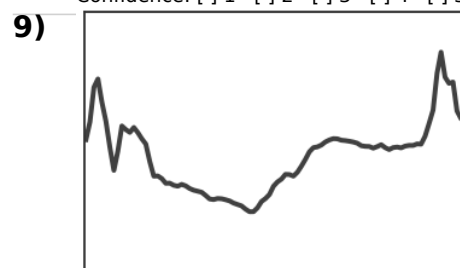
Label: ☐ Red ☐ Blue

Confidence: ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5



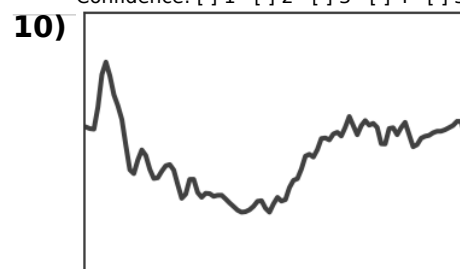
Label: ☐ Red ☐ Blue

Confidence: ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5



Label: ☐ Red ☐ Blue

Confidence: ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5



Label: ☐ Red ☐ Blue

Confidence: ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

Please answer immediately after finishing this task.

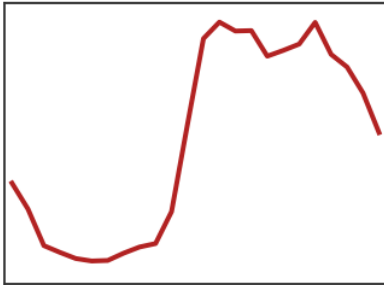
| Statement                                              | 1 Strongly disagree      | 2 Disagree               | 3 Neutral                | 4 Agree                  | 5 Strongly agree         |
|--------------------------------------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| I understood the difference between the classes.       | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| The information shown was sufficient.                  | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| The explanation was easy to apply.                     | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| The explanation matched visible patterns in the plots. | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| I felt confident in my classifications.                | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| The explanation was too vague.                         | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| The explanation contained too much detail.             | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |

Optional note about Task 1:

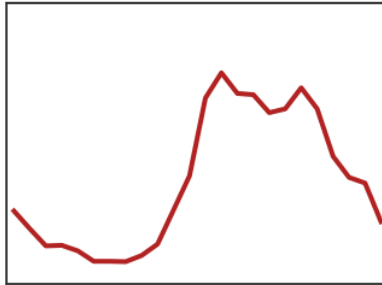


Study the example plots and the short reference notes below. Then classify the 10 test instances on the next pages.

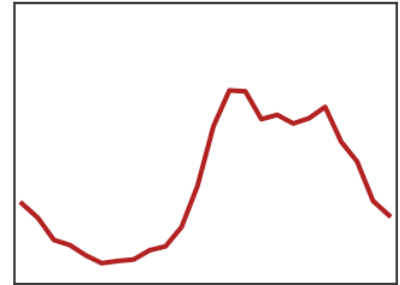
### Red reference examples



Example 1



Example 2



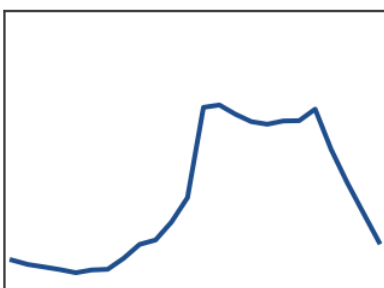
Example 3

#### Reference notes and rules

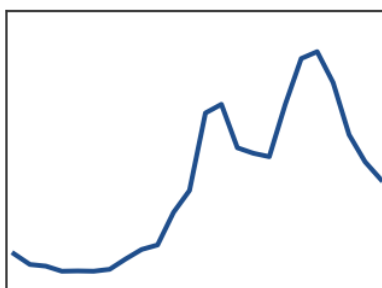
R1: In the early region, average values are clearly above zero (around 0.1 or higher); the series starts at a moderate level before dipping.

R2: Within the early region, there is a strong downward drop (the first value is at least about 0.15 higher than the early minimum), giving a pronounced descending shoulder.

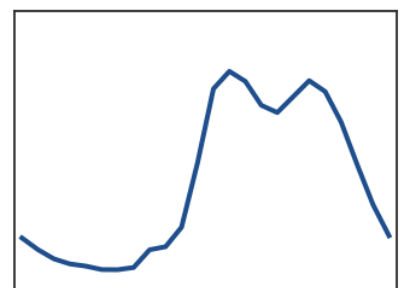
### Blue reference examples



Example 1



Example 2



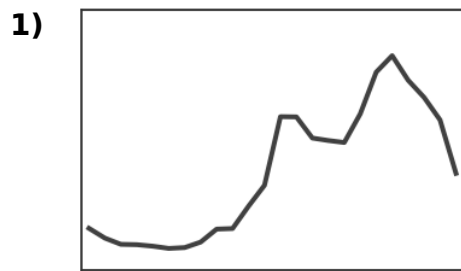
Example 3

#### Reference notes and rules

R1: In the early region, average values stay very close to zero (below about 0.1); the series begins on a low baseline.

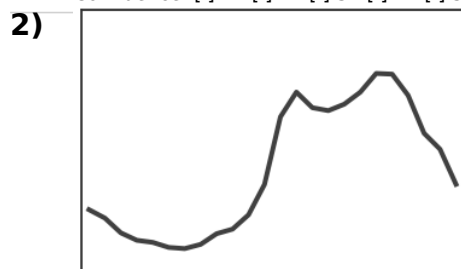
R2: Within the early region, variation is small (the difference between highest and lowest early values is less than about 0.15), so this part appears almost flat near zero.

For each plot, choose Red or Blue and then rate your confidence from 1 (very low) to 5 (very high).



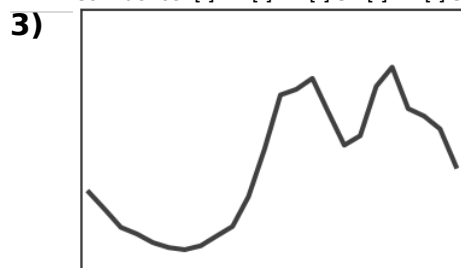
Label: ☐ Red ☐ Blue

Confidence: ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5



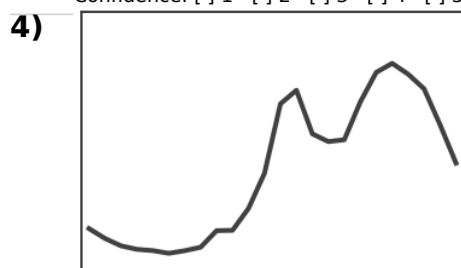
Label: ☐ Red ☐ Blue

Confidence: ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5



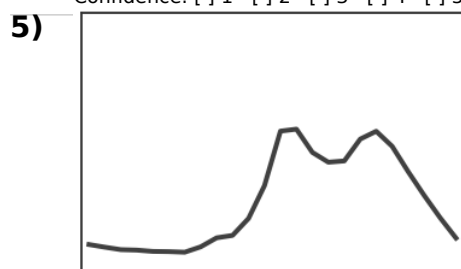
Label: ☐ Red ☐ Blue

Confidence: ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5



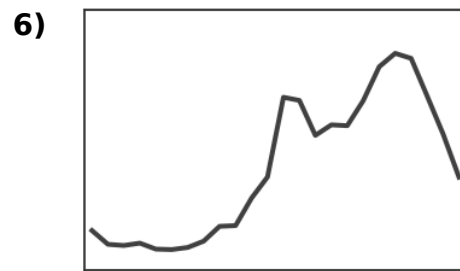
Label: ☐ Red ☐ Blue

Confidence: ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5



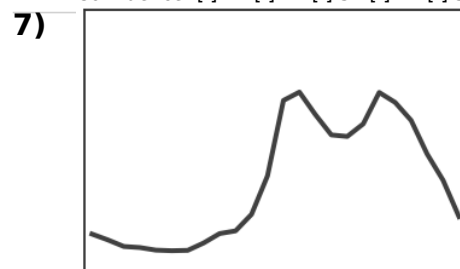
Label: ☐ Red ☐ Blue

Confidence: ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5



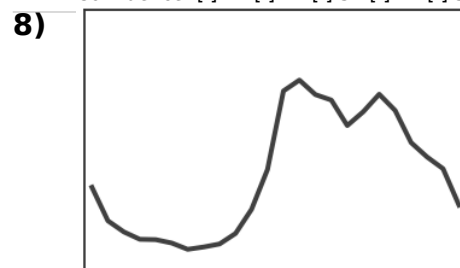
Label: ☐ Red ☐ Blue

Confidence: ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5



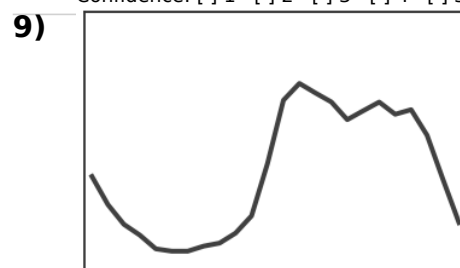
Label: ☐ Red ☐ Blue

Confidence: ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5



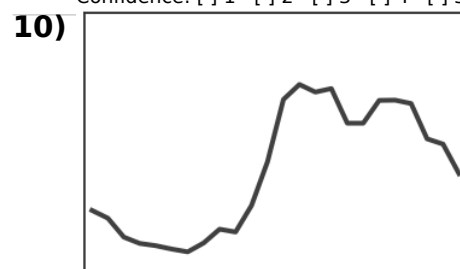
Label: ☐ Red ☐ Blue

Confidence: ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5



Label: ☐ Red ☐ Blue

Confidence: ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5



Label: ☐ Red ☐ Blue

Confidence: ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5



Please answer immediately after finishing this task.

| Statement                                              | 1 Strongly disagree      | 2 Disagree               | 3 Neutral                | 4 Agree                  | 5 Strongly agree         |
|--------------------------------------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| I understood the difference between the classes.       | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| The information shown was sufficient.                  | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| The explanation was easy to apply.                     | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| The explanation matched visible patterns in the plots. | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| I felt confident in my classifications.                | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| The explanation was too vague.                         | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| The explanation contained too much detail.             | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |

Optional note about Task 2:

Use these prompts after both tasks.

1. What visual features did you use to classify?
2. In the task where rules were shown, did the rules help? If yes, which part?
3. Were any rules confusing, vague, or hard to see in the plots?
4. Did you rely more on prototypes or rules in the rules condition?
5. Would you prefer prototypes only, rules only, or both?
6. Did the rules make you understand the classifier better, or mainly help answer the test?

## Interviewer notes: